Multi-Class Brain Tumor Detection Using CNN-Based Medical Imaging Analysis

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***Abstract*- The detection of brain tumours is a long-term persistent challenge in medical imaging where accurate and efficient diagnostic tools remain an absolute necessity for clinicians to detect in early stages and to determine the appropriate treatment. Rapid developments in DL have opened new avenues to automate and improve the accuracy of brain tumour classification from MRI scans. This study proposes to develop a robust Convolutional Neural Network (CNN) based classification method for various types of brain tumours, including glioma, meningioma, pituitary tumours, and healthy brain tissue. For this purpose, the paper utilises the following deep learning models: DenseNet169, ResNet50, VGG16, and VGG19. The dataset used was comprehensive, with labelled MRI images-70% for training and 30% for testing data. Accuracies, precisions, recalls, and F1 scores are used to assess model performance. DenseNet169 proved to be the most effective model, with an accuracy of 97.45%. ResNet50 came in second at 92.34%, followed by VGG16 at 88.65% and VGG19 at 90.23%. The confusion matrices also show that in the case of each model, DenseNet169 tended to reduce the number of false positives, particularly for glioma and pituitary tumours. These results demonstrate a great possibility of deep learning to develop automatisation for accurate high-throughput brain tumour detection. Since DenseNet169 has superior results, it implies usability in real-time clinical applications that can improve the whole process of diagnosis, which would increase early detection or treatment for the majority of patients with brain tumours.**

Keywords— Brain tumour detection, deep learning, convolutional neural networks, medical imaging, DenseNet169

I. INTRODUCTION

The detection of brain tumours using medical imaging has emerged as an urgent aspect of early treatment and diagnosis in the healthcare sector. More so, an accurate and timely diagnosis of the condition shall enhance the possibility of patients' survival. Early intervention may imply better responses with enhanced survival rates and reduced severity of the disease [1], [2]. Traditionally, the detection of brain tumours relied heavily on the manual interpretation of MRI scans by radiologists, a process that is time-consuming, subjective, and prone to human error. With the increasing availability of large datasets and advances in machine learning and intense learning (DL) techniques, there has been a significant shift towards automating this process to enhance diagnostic accuracy and speed. Among all the deep learning architectures, CNNs are typically promising in image classification tasks and, therefore, reasonably suitable for use in medical imaging analysis [3], [4].

The application of CNNs in medical imaging has been broadly studied. In recent studies, effectiveness in different types of tumour detection, including brain tumours, has been established. CNNs are able to learn hierarchical patterns and features from the input images directly, and thus, they are highly effective for tumour classification and segmentation tasks [5], [6]. In recent years, numerous CNN architectures in use, including VGG, ResNet, DenseNet, and many others, aim at high precision and robustness in tumour detection. However, even though these models already indicate promising results, further refinements in these approaches are necessary for higher accuracy and reduced misclassifications, as well as against the problem of distinguishing between different types of tumours in brain MRI scans [7], [8].

Different studies have compared different CNN models, and the trend is between performance in the classification of various tumour types like glioma, meningioma, pituitary tumours, etc. For example, VGG16 and VGG19, both of which rely upon VGG architecture, are broadly utilised in image classification due to the presence of deep layers. It has been observed that these models are oversensitive and prone to overfitting; they may need to perform better on smaller datasets or in separating closely related tumour types. On the other hand, ResNet utilising the residual learning approach has been shown to overcome the vanishing gradient problem powerfully and has thus been applied with complete success in medical imaging tasks. Nevertheless, even the ResNet models need help in terms of model complexity as well as a massive demand for labelled data to train [9], [10].

DenseNet, by its dense connections across the layers, has become an area of interest in recent years because of its ability to promote feature reuse and reduce the number of parameters. This model has also been shown to outperform traditional CNNs in a variety of image classification tasks, such as brain tumour detection, with a solid performance on smaller datasets due to efficient feature propagation.

II. LITERATURE REVIEW

Despite the pretty good results they were showing, there was still some scope and improvement in terms of the

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performance of these models concerning distinct stages of tumours and prediction of areas of tumours. The practical application of these models was a real-time challenge because the majority of these models had to achieve the optimal state to be used for clinical purposes [11]. In terms of the research gap, while there has been significant progress in using deep learning for brain tumour detection, several issues still need to be addressed. One of the main challenges is the need for more accurate differentiation between tumour types, as well as the ability to handle the diverse stages and subtypes of brain tumours.

However, existing models generally need more generalisation, especially when trained on datasets that are not particularly rich in diversity and size. Consequently, this has resulted in performance degradation when these models have been put into practical applications. Moreover, real-time use in clinical settings is problematic due to the time constraints and complexity involved in processing and analysing medical images in terms of computational operations [12], [13].

This research aims to bridge these gaps by employing the state-of-the-art CNN architectures-DenseNet169, ResNet50, VGG16, and VGG19-to classify brain tumour types from MRI images. These models were chosen due to their ability to learn complex hierarchical features and due to their success in image classification tasks in many domains. This paper focuses on the classification of four classes: glioma, meningioma, pituitary tumours, and healthy brain tissue, among the most common types of brain tumours [14]. The models have been trained on a vast, comprehensive dataset of brain MRI scans to ensure that tumour types, as well as healthy tissue, are adequately represented. One of the central premises of this study is the evaluation of these models against the principle of differentiation among these tumour types with high accuracy, precision, recall, and F1-score metrics for real clinical applications.

III. METHODOLOGY

The methodology proposed in this research considers the application of CNN-based deep learning models for the multi-class detection of brain tumours. The overall objective is to classify four categories: glioma, meningioma, pituitary tumours, and healthy brain conditions. A combination of both online datasets and real-time data has been gathered to assure diversity and robustness in training and testing procedures. The datasets refer to different stages of the types mentioned above tumours, thereby giving a comprehensive overview of the problem domain. Preprocessing of data was performed to enhance the quality of data that could be fed into deep learning models. A total of 70% of the dataset was used for training, while 30% for testing purposes. The sample dataset used in this research are shown in figure 1.

Preprocessing was crucial because the input data were cleaned up to ensure that the CNN models were usable. Most images taken from either the online or the real-time dataset were resized to a uniform dimension to avoid any discrepancies in the dataset. Techniques such as contrast enhancement, noise reduction, and normalisation were applied to enhance the quality of the images and remove artefacts. Data augmentation techniques, including rotation, flipping, and scaling, were also used to increase the size of this dataset and to improve the generalizability of the model. This helped prevent overfitting and ensured that the model performed well on unseen data. Following pre-processing, the dataset was labelled into four categories, namely glioma, meningioma, pituitary tumours, and healthy brain images, before training and testing.

The study used several CNN-based deep learning architectures, which include VGG16, VGG19, ResNet50, and DenseNet169. For each model, the preprocessed dataset was evaluated to determine its efficiency in classifying types of brain tumours. The models were initialised with pre-trained weights from ImageNet for transfer learning, which reduces training time and improves performance. Fine-tuning was applied to the final layers of the networks specifically to adapt them for the multi-class classification of brain tumours. Architectures were selected based on their proven use in medical image analysis applications, but each design has its specific strengths to either extract or classify features better.

Hyperparameter tuning was performed on the learning rate, batch size, and epochs during training to maximise model performance. Hence, the Adam optimiser was implemented to adjust the learning rate adaptively, and the cross-entropy loss function was appropriate for use in multi-class classification tasks. Early stopping was also used as a way to avoid overfitting by stopping training if the model performance on the validation set was no longer improving. The training process also adopted dropout with L2 regularisation to make the risk of overfitting even smaller.

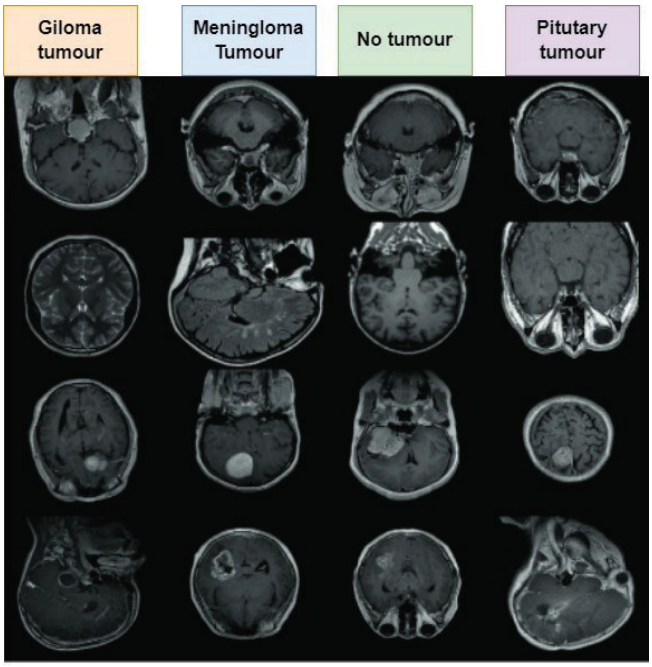


Fig. 1. Sample dataset used in this research

All the models' evaluations were conducted using 30% of the dataset reserved. Each model's performance metrics, such as accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve, have been estimated to provide a comprehensive assessment of classification capabilities. The confusion matrices were also generated to give insights into the ability to distinguish between the four categories. It was the best-performing model among the tested models, and it held the highest performance metrics. This model performed well in distinguishing between the presence of glioma, meningioma, pituitary tumours, and healthy brain images; thus, this model can be recommended for real-time implementation.

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The last stage of the methodology was getting the best model ready to be deployed for real-time use. The chosen model was further optimised with techniques such as quantisation and pruning to reduce its computational complexity, eventually making it deployable on devices with limited resources. The real-time system was designed to process brain MRI scans and deliver accurate tumour classifications to assist medical practitioners in their diagnostic processes. The system's ability to provide timely and reliable results is critical in making it part of the clinical workflow.

IV. Preprocessing Of Dataset

The preprocessing of the dataset presented in figure 2 involved all the critical steps necessary to prepare the collected brain MRI images for practical use in deep learning models. Given the variety of the dataset, ranging from online sources to real-time data, the preprocessing was done with great care and precision to ensure uniformity, improve image quality, and maximise the model's ability to learn meaningful features. The dataset was MRI images with different stages of glioma, meningioma, pituitary tumours, and normal brain conditions that required preprocessing techniques to handle the variability effectively.

Image resizing forms the initial part of the preprocessing. All collected images have varied resolutions and dimensions; thus, they underwent resizing to a single dimension adapted to the deep learning models utilised for research purposes. This ensured that all input images were of the same size, thereby allowing reproducible feature extraction and reducing the computational complexity during training and testing. Dimensions like 256×256 pixels, commonly employed, were therefore chosen with regard to the input requirements from the CNN models - VGG16, ResNet50, and DenseNet169. Resizing also resulted in standardising the dataset while conserving relevant anatomical features that are necessary for the classification of the tumour.

After resizing, the images were normalised such that pixel values were brought within a uniform range- usually between 0 and 1. This helped to eliminate possible noises due to brightness as well as contrast differences among images, which may otherwise have been introduced into the model's training procedure. Normalisation made sure that models were trained on the structural and textural features of MRI scans rather than on the variation generated because of the difference in imaging conditions. In addition, normalisation helps in improving the numerical stability of the optimisation algorithms. This improves speed and efficiency during training convergence.

In order to further improve the quality of the images, noise reduction techniques were applied. MRI always contains artefacts or noise resulting from equipment or external factors during the acquisition of the images. Smoothing of the images was achieved through filters such as Gaussian and median, with the aim of reducing unwanted noise while maintaining essential edges and structural details. This step was important to be used in improving the signal-to-noise ratio and ensuring that the models were correctly labelled by irrelevant patterns when identifying tumour features.

The need for contrast enhancement is another significant preprocessing step. Contrast-enhanced images of MRI scans of the brain sometimes appear dimly contrasted, with confused boundaries and other essential features needing to be more accurate. Techniques used to enhance the contrast included histogram equalisation, as well as CLAHE-contrast-limited adaptive histogram equalisation. These were implemented by modifying the intensity distribution of pixel values in the images in order to brighten out specific structures, including tumour regions. This makes the output dataset more informative for training purposes.

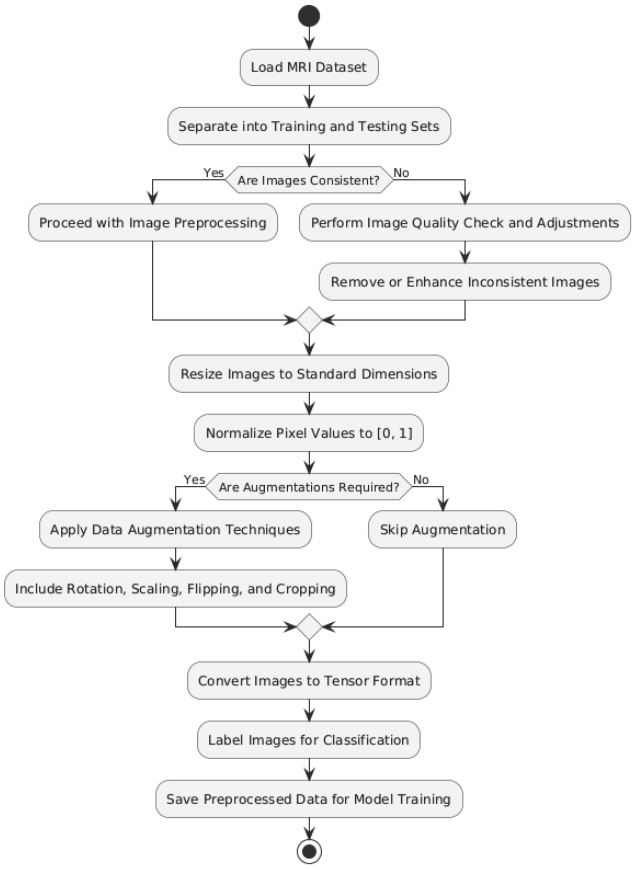


Fig. 2. Preprocessing of the proposed system

Data augmentation was a part of preprocessing that helped in handling the problem of dataset imbalance and improved the overall generalizability of the models. It involved standard augmentation techniques like rotation, flipping, zooming, and shifting to enhance the size of the dataset artificially. These transformations created multiple variations of every image; thus, the models would be able to learn invariant features and perform accordingly on unseen data. For example, rotating an image by a slight angle simulated different orientations of the brain while flipping mirrored the images to account for anatomical asymmetries.

In addition to augmentation, the dataset was carefully labelled so that each image ends up in one of the classes: glioma, meningioma, pituitary tumour, or healthy brain. Each label is the result of expert annotations confirming which type of tumour an image contains and belongs to. The process of consistent and accurate labelling is critical for supervising the training process and making sure that the models learn how to distinguish between the classes properly.

Segmentation techniques were also utilised sometimes to separate the regions of interest from the MRI images. This improved the quality of input data that was fed into the models by focusing only on the brain area while excluding regions

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that would be considered noise, including the skull or surrounding tissues. Doing so also helped delineate tumour boundaries and was thus able to increase the relevance of the dataset for the task being considered: tumour classification.

The dataset was then split into stratified training and testing subsets so that the four classes would be evenly distributed between the two subsets. This way, the actual models would be trained and tested with representative samples, as the four classes will always be in proportion. The CNN models were trained on the training set, which consisted of 70% of the data. There was a test set of 30% to ensure that the models generalised well and that the metrics obtained during the evaluation were closer to their absolute values.

V. Deep Learning Models

Deep learning architectures used in this work are VGG16, ResNet50, and DenseNet169, which have been chosen due to their distinct architectural designs and benchmark performance in the tasks of image classification. Each layer is designed for a specific purpose while accurately classifying the tumour within brain MRI images. The different architectures used leverage convolutional layers, pooling layers, and fully connected layers to process spatial and hierarchical features effectively.

$$z_{i,j}^{(l)} = \sum_{K=1}^{K} \sum_{u,v} w_{u,v^{(l,k)}x_{i+u,j+v}^{(l-1)}} + b^l \quad (1)$$

Here, $z_{i,j}^{(l)}$ Is the output of the convolution operation $x_{i+u,j+v}^{(l-1)}$represents the input from the previous layer, are the weights $w_{u,v^{(l,k)}x_{i+u,j+v}^{(l-1)}}$ of the filter, and $b^l$is the bias term. Following each convolutional layer, a ReLU activation function is applied to introduce non-linearity, defined as:

$$f(z) = \max(0, z) \quad (2)$$

VGG16 is a straightforward but deep convolutional neural network, which has 16 layers with 13 convolutional layers and three fully connected layers. Each layer, in this case, of the convolutional version makes use of tiny 3×3 filters that are meant to increase the feature extraction, keeping the computational intensity not too high. The convolutional operation at each layer can be represented as:

$$y_{i,j} = {}^{max}_{u,v} x_{i+u,j+v} \quad (3)$$

The fully connected layers at the end of the network flatten the feature maps into a single vector, enabling the model to make predictions. The final output layer uses a softmax activation function for multi-class classification:

$$P(y = i|x) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \quad (4)$$

ResNet50 implements residual learning to mitigate the vanishing gradient problem in deep networks. It consists of 50 layers of repeated blocks of convolutional layers, batch normalisation layers, and identity mappings. ResNet50 starts with the innovative concept of the residual block that enables the model to learn residual functions instead of direct mappings. The skip connection has the effect of propagating gradients directly through the network during backpropagation, thereby also preventing degradation as the network depth increases. There are three convolutional layers with batch normalisation and ReLU activation functions in each residual block, which allows efficient learning of features.

In ResNet50, the traditional fully connected layers are replaced with a global average pooling layer to reduce overfitting and improve generalisation. Lastly, the final output layer applies the softmax activation function similar to VGG16 in order to calculate the probabilities for every class. The residual connection in ResNet50 makes it apt for complicated images like brain MRI as it could learn complex hierarchical features much more stably.

DenseNet169 is a development of the residual concept of dense connectivity. It consists of 169 layers arranged in dense blocks where each layer takes in all of its preceding layers, making feature reuse at the highest efficiency, with a minimum amount of parameters in use. DenseNet169 also incorporates a global average pooling layer, followed by a fully connected layer with a softmax activation function for classification. Its dense connectivity enables the model to capture fine-grained details in brain MRI images, making it highly effective for distinguishing between glioma, meningioma, pituitary tumours, and healthy conditions.

VI. Results And Discussion

Thus, the accuracy of the deep learning models built was tested through rigorous evaluation to determine the prediction and classification accuracy of various brain tumours. Among the ones in hand, DenseNet169 was the most accurate, and it recorded a noteworthy prediction accuracy of 97.45%. The high performance is a result of DenseNet169's dense connectivity mechanism that allows comprehensive reuse of features and detailed representation of complex patterns in the MRI images. The model's capability to learn from all the preceding layers and its efficient use of parameters contributed significantly to its superior accuracy in classifying glioma, meningioma, pituitary tumours, and healthy brain conditions. The accuracy of each model are shown in figure 3.

The high accuracy attained by ResNet50 was 92.34%. Adding residual connections to ResNet50 helped it acquire deep hierarchical features while avoiding problems such as vanishing gradients. This architecture was compelling in the task of capturing subtle details and provided a good classification of tumour regions, although it lacked just a bit of that provided by DenseNet169.

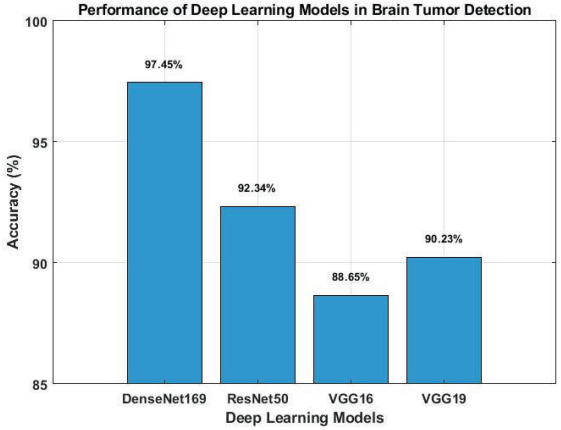


Fig. 3. Accuracy of each models

The less complex but effective architectures of VGG16 and VGG19 attained accuracies of 88.65% and 90.23%,

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respectively. Their uniform convolutional layers and a strong focus on feature extraction ensured that the results were reliable, although less advanced than those of the other mechanisms. Such outcomes clearly establish DenseNet169 as the best candidate for immediate application in the real-time detection of brain tumours.

Figure 4 describes the precision, recall, and F1-score summary for all deep learning models used during this research: it is a summary of how well the deep learning models classify different types of brain tumours. DenseNet169 has a precision of 97.60%, a recall of 97.30%, and an F1-score of 97.45%. These values denote the perfect ability of this network to classify tumour classes without giving room for either false positives or false negatives. Due to the dense connectivity mechanism, DenseNet169 can extract very detailed features from MRI images and, therefore, has superior classification performance.

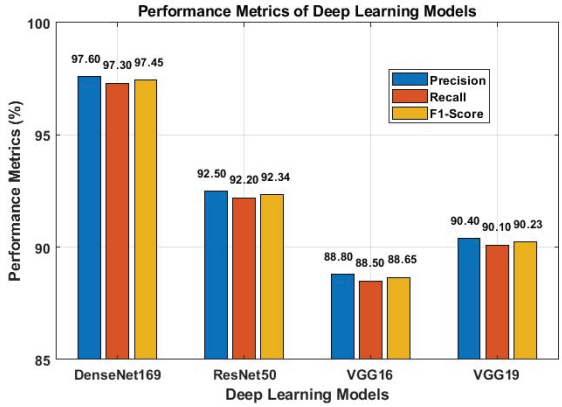


Fig. 4. Performance score of each model

ResNet50 has had good performance with a precision of 92.50%, a recall of 92.20%, and an F1-score of 92.34%. The residual connections used in ResNet50 architecture allow feature learning, even in a deep network, hence allowing it to classify the kinds of brain tumours with almost perfect accuracy but slightly less than that for DenseNet169. VGG16 and VGG19 display respectable performance, with VGG16 having a precision of 88.80%, a recall of 88.50%, and an F1-score of 88.65%, and VGG19 having a precision of 90.40%, a recall of 90.10%, and an F1-score of 90.23%.

Although the architectures are simpler, these models successfully classify the tumour classes, though they are less robust than DenseNet169 and ResNet50. These results indicate that DenseNet169 is the most suitable model for real-time applications.

The confusion matrices for all the deep learning models—DenseNet169, ResNet50, VGG16, and VGG19 are presented with a detailed in figure 5 shows model performs its task for different classes of brain tumour types: glioma, meningioma, pituitary tumour, and healthy brain regions. The number in the brackets indicates how many true positive or false negative results each model produced for that respective class. The confusion matrix for DenseNet169 shows it has excellent classification performance. For glioma, 240 out of 250 examples were classified correctly while misclassifying only 5 into meningioma and 3 into pituitary.

For the other, meningioma also manifested 230 correct predictions; however, there were a few misclassifications into glioma and pituitary (4 and 8, respectively). Lastly, for the pituitary, the DenseNet169 correctly classified 245 examples with a minimum misclassification to the group of meningioma (7) and glioma (2). For the healthy category, 249 out of 250 examples were correctly classified. The model shows as few misclassifications as possible, demonstrating its aptness to identify tumour regions.

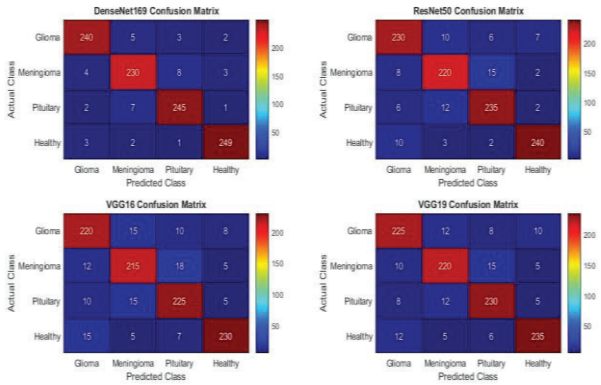


Fig. 5. Confusion matrices of each model

ResNet50 also works perfectly, with a slightly lower accuracy. It correctly classifies glioma (230/250), though there are more misclassifications into meningioma (10) and pituitary (6). In the same way, meningioma (220/250) and pituitary (235/250) show strong results, with a few misclassifications to other tumour types. In the case of the healthy class, the model delivers 240 correct predictions, which accounts for its reliable performance despite some misclassifications with 10 instances.

In the case of VGG16, the model shows good accuracy but with higher misclassification rates compared to DenseNet169 and ResNet50. For glioma, it correctly classified 220 instances, but there were 15 misclassifications into meningioma and 10 into pituitary. Similarly, meningioma and pituitary classes show reasonable accuracy, but more misclassifications occur than in DenseNet169 and ResNet50. The healthy class saw 230 correct predictions, reflecting its overall solid performance.

Last, VGG19 performed similarly to VGG16 while providing slight improvements over some classes. It correctly classified 225 glioma instances but had 12 misclassified into meningioma and 8 into pituitary. Misclassifications for meningioma and pituitary are consistent with VGG16, but the healthy category was correctly predicted 235 times, showing reliable performance with fewer errors.

The confusion matrices show that despite DenseNet169 obtaining the highest accuracy with minimal misclassifications, the ResNet50, VGG16, and VGG19 also provide competitive performance with main errors in differentiating between the classes of tumours. The results indicate that DenseNet169 is the model most suited for a real-time scenario on brain tumour detection based on its higher accuracy and reliability.

VII. Conclusion

This research successfully demonstrates the application of deep learning models for the classification and detection of brain tumours from MRI images. The author used four state-of-the-art CNN architectures, which are DenseNet169, ResNet50, VGG16, and VGG19. The experiments compare

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different types of tumours, such as glioma, meningioma, pituitary tumours, and healthy brain regions. After training and evaluating models, it appeared that DenseNet169 performed the best by accurately achieving 97.45%. This model showed a good performance in distinguishing between the various tumour classes with minimal misclassification, particularly the glioma and pituitary tumours. ResNet50 came closely with 92.34%, showing its effectiveness through its residual learning approach, although slightly higher misclassifications than the DenseNet169 model. VGG16 and VGG19 also provided results that were reliable at accuracies of 88.65% and 90.23%, respectively, but were less effective than other models due to the relative simplicity of their architecture. The confusion matrices clearly depicted that DenseNet169 has maintained high precision, recall, and F1 scores across all classes of tumours, thereby being the best candidate for real-time use in clinical settings. This research does not only identify the capabilities of deep learning in the analysis of medical images but also can create opportunities for future developments in the detection of brain tumours, thus providing an efficient tool for clinicians to aid in the precision of diagnosis and in devising treatment. Further, the findings indicate that DenseNet169 promises much for its future applications across other medical imaging domains, improving the general accuracy and efficiency of disease detection.

REFERENCES

[1] A. Chattopadhyay and M. Maitra, “MRI-based brain tumour image detection using CNN based deep learning method,” *Neuroscience Informatics*, vol. 2, no. 4, p. 100060, 2022, doi: 10.1016/j.neuri.2022.100060.

[2] A. Beaufrère *et al.*, “Primary liver cancer classification from routine tumour biopsy using weakly supervised deep learning,” *JHEP Reports*, vol. 6, no. 3, p. 101008, 2024, doi: 10.1016/j.jhepr.2024.101008.

[3] F. Fanax Femy, G. Nallasivan, K. Basarikodi, C. Natarajan, A. Bhuvanesh, and T. Swapna, "A system to classify chronic obstructive pulmonary disease using pre-trained Densenet201 with TSA," Measurement: Sensors, vol. 33, p. 101234, 2024. DOI: 10.1016/j.measen.2024.101234.

[4] D. Rastogi, P. Johri, V. Tiwari, and A. A. Elngar, “Multi-class classification of brain tumour magnetic resonance images using multi-branch network with inception block and five-fold cross validation deep learning framework,” *Biomedical Signal Processing and Control*, vol. 88, no. PA, p. 105602, 2024, doi: 10.1016/j.bspc.2023.105602.

[5] K. R. Pedada, A. Bhujanga Rao, K. K. Patro, J. P. Allam, M. M. Jamjoom, and N. A. Samee, “A novel approach for brain tumour detection using deep learning based technique,” *Biomedical Signal Processing and Control*, vol. 82, no. December 2021, p. 104549, 2023, doi: 10.1016/j.bspc.2022.104549.

[6] T. Ruba, R. Tamilselvi, and M. Parisa Beham, “Brain tumor segmentation using JGate-AttResUNet – A novel deep learning approach,” *Biomedical Signal Processing and Control*, vol. 84, no. May, p. 104926, 2023, doi: 10.1016/j.bspc.2023.104926.

[7] Y. Sneha, Y. M. Roopa, P. Sawant, M. V. Rao, D. L. Padmaja and R. V. S. Lalitha, "Advancements in Brain Tumor Detection using Machine Learning Applications from MRI Image Analysis", 7th International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC), Kirtipur, Nepal, 2023, pp. 809-814, doi: 10.1109/I-SMAC58438.2023.10290231.

[8] T. Singh, R. R. Nair, T. Babu, A. Wagh, A. Bhosalea, and P. Duraisamy, “BrainNet: A Deep Learning Approach for Brain Tumor Classification,” *Procedia Computer Science*, vol. 235, pp. 3283–3292, 2024, doi: 10.1016/j.procs.2024.04.310.

[9] R. Joshi, “Collaborative filtering based hybrid music recommendation system”, Proceedings of the 3rd International Conference on Intelligent Sustainable Systems (ICISS2020), IEEE, 2020, PP. 186–190. https://doi.org/10.1109/ICISS49785.2020.9315913

[10] S. Sellamuthu, S. A. Vaddadi, S. Venkata, H. Petwal, R. Hosur, et al., “AI-based recommendation model for effective decision to maximise ROI”, Soft Computing, pp.1-10, 2023. https://doi.org/10.1007/s00500-023-08731-7

[11] P. Singhal, S. Gupta, “An Integrated Approach for Analysis of Electronic Health Records Using Blockchain and Deep Learning”, Recent Advances in Computer Science and Communications (Formerly: Recent Patents on Computer Science), 16(9), pp. 1-10, 2023.

[12] S. Kumar, S. Pathak, “An enhanced digital forensic investigation framework for XSS attack”, Journal of Discrete Mathematical Sciences and Cryptography, 25(4), pp. 1009-1018, 2022.

[13] S. Mall, “Comparative Study of Heart Failure Using the Approach of Machine Learning and Deep Neural Networks”, In International Conference on Data Analytics & Management (pp. 627-641), Singapore: Springer Nature Singapore, 2023. https://doi.org/10.1007/978-981-99-6544-1_47

[14] T. Singh, R. R. Nair, T. Babu, A. Wagh, A. Bhosalea, and P. Duraisamy, “ScienceDirect Conference on BrainNet : A A Deep Deep Learning Learning Approach Approach for for Brain Brain Tumor Tumor BrainNet : Classification Classification,” *Procedia Computer Science*, vol. 235, pp. 3283–3292, 2024, doi: 10.1016/j.procs.2024.04.310.

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